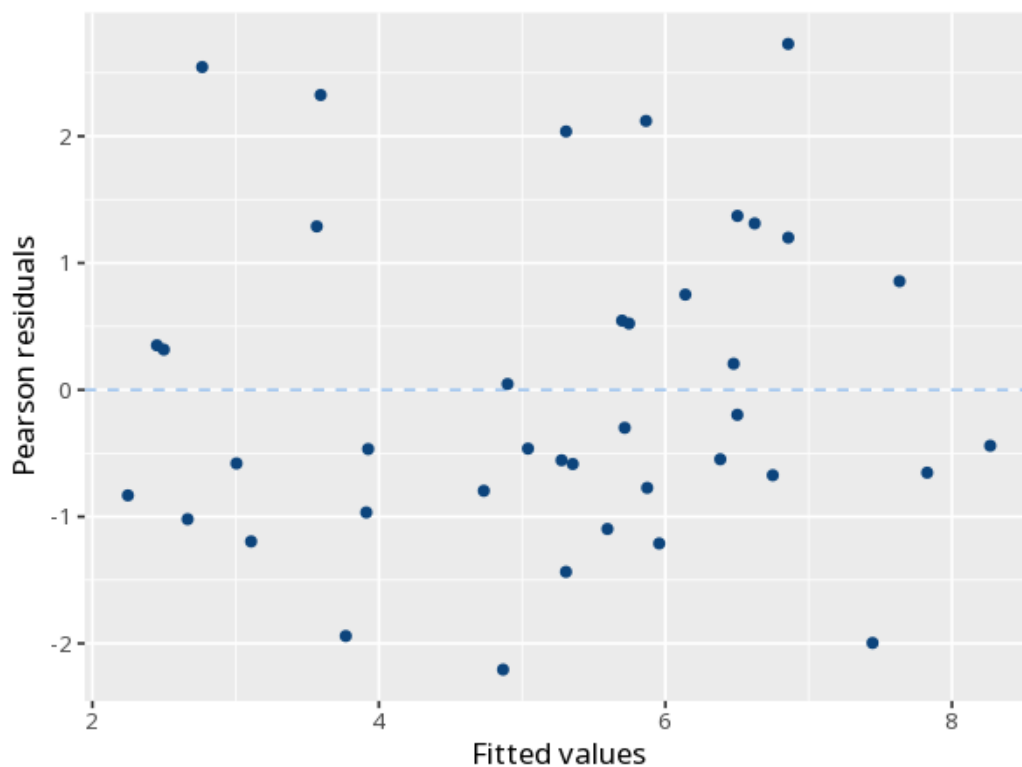


1 Poisson / negative binomial

	Log-count (B)	IRR
(Intercept)	-1.46 (0.23)	0.23 [0.15, 0.36]
age	0.01 (0.005)	1.01 [1.00, 1.02]
group: b	0.17 (0.14)	1.19 [0.90, 1.56]
Num.Obs.	40	
Dispersion	1.69	
Residual deviance	67.57	
df	37	
Log.Lik.	-98.18	
AIC	202.37	
BIC	207.44	



Models event counts. Each predictor's incidence-rate ratio (IRR) gives the multiplicative change in the expected count per unit (>1 more, <1 fewer); p tests significance. Check dispersion to pick Poisson vs. negative binomial. If

cases have unequal exposure (different observation time/area/population), add an offset of $\log(\text{exposure})$ so the model predicts rates, not raw counts — omitting it biases the IRRs. Your significance threshold (α) is 0.05.

Predictor age was associated with the count, IRR=1.01, 95% CI [1.00, 1.02], $p = .007$.

Statistical basis: Poisson regression (incidence-rate ratios) (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Negative binomial regression (glm.nb) for overdispersed counts (Venables WN, Ripley BD (2002). Modern Applied Statistics with S, Fourth edition. Springer, New York.); Overdispersion check (Lüdecke D, Ben-Shachar MS, Patil I, Waggoner P, Makowski D (2021). "performance: An R Package for Assessment, Comparison and Testing of Statistical Models." Journal of Open Source Software, 6(60), 3139.).